

1. For each equation below, write the letter that represents the best name of that 3D figure. [1 pt each]

- | | | |
|----------------------------------|-----------------------------------|---------------------------------|
| A: Line | B: Two lines | C: Ellipsoid |
| D: Plane | E: Two planes | F: Elliptic Cone |
| G: Hyperboloid of 1 Sheet | H: Hyperboloid of 2 Sheets | I: Hyperbolic Paraboloid |
| J: Elliptic Paraboloid | K: Elliptic Cylinder | L: None of the above |

$$y^2 - x^2 = z$$

$$y^2 - x^2 - z = 0$$

I

$$\frac{x^2}{4} + y^2 - \frac{z^2}{9} = 0$$

F

$$z = 2x + 3$$

D

$$\frac{x^2}{4} + y^2 - \frac{z^2}{9} = 1$$

G

$$x^2 + z^2 = y - 1$$

$$x^2 + z^2 - y = -1$$

J

$$\frac{x^2}{4} + y^2 - \frac{z^2}{9} = -1$$

H

2. Consider the hyperbolic paraboloid $y^2 - z^2 = -x$. Write the letter that best represents each trace below. [1 pt each]

- S: parabola that opens in the direction of the positive x-axis
 T: parabola that opens in the direction of the negative x-axis
 U: hyperbola that opens in the direction of the y-axis
 V: hyperbola that opens in the direction of the z-axis
 W: two intersecting lines

i) Trace at $y = 0$ is S

ii) Trace at $z = 0$ is T

iii) Trace at $x = 0$ is W

iv) Trace at $x = 1$ is V

$$y^2 = z^2$$

$$y = z$$

$$-y = z$$

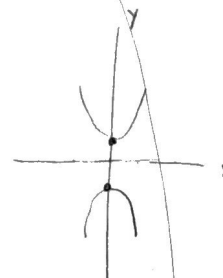
$$y^2 - z^2 = -1$$

$$z^2 - y^2 = 1$$

$$y^2 - z^2 = 0$$

$$y^2 - z^2 = -1$$

$$x^2 - y^2 = -1$$



$$x^2 - y^2 = 0$$

-0

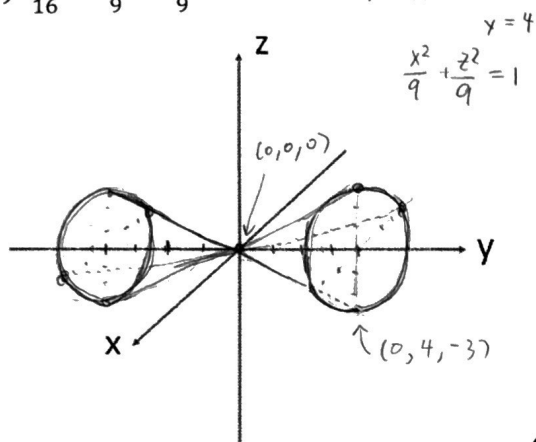
3. Sketch each 3D figure below. Make sure that your sketch clearly includes:

- traces in at least 2 directions should be visible
- one key point (vertex / intercept / saddle point) labeled with its coordinates
- Then, state the name of each figure.

[3 pts each sketch, 1 pt name]

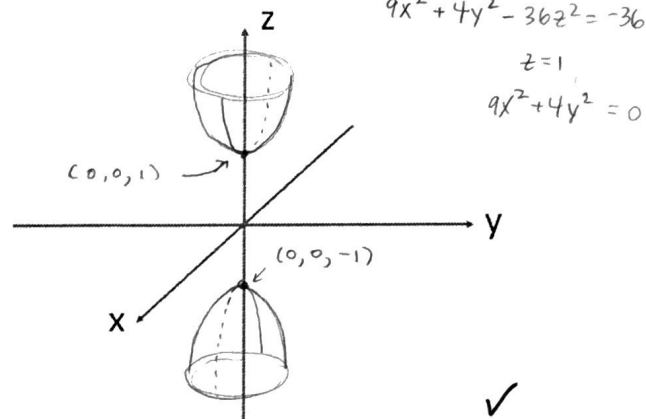
a) $\frac{y^2}{16} = \frac{x^2}{9} + \frac{z^2}{9}$

$\frac{x^2}{9} + \frac{z^2}{9} - \frac{y^2}{16} = 0$



Name: elliptic cone

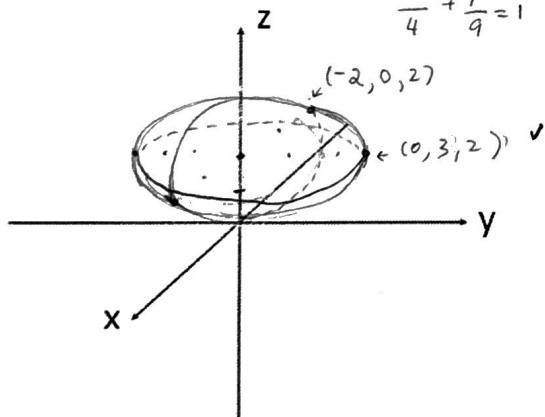
b) $36z^2 - 4y^2 - 9x^2 = 36$



Name: hyperboloid of 2 sheets

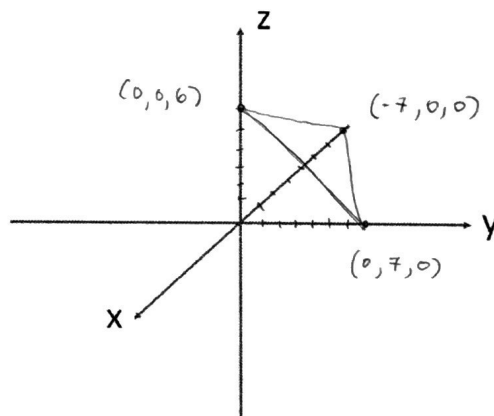
c) $\frac{x^2}{4} + \frac{y^2}{9} + (z-2)^2 = 1$

$z=2$
 $\frac{x^2}{4} + \frac{y^2}{9} = 1$



Name: ellipsoid

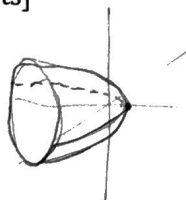
d) $-6x + 6y + 7z = 42$



Name: plane

4. Write an equation of an elliptic paraboloid with vertex at (0,1,0) that opens along the negative y-axis.
[3 pts]

-1



$x^2 + z^2 - y = -1$

This opens
the other way

CO (not O)

$y \leq 1$

5. Enlighten your Analysis Honors teachers, please. What is the "six-seven" thing about? What does it mean

TO YOU? [0 pts]

I don't rly

basically it originated from a rap song with "six-seven" in it → it became

a meme and now ppl just like it since "six-seven" are numbers so common

You can bring it up a lot, it's supposed to be nonsensical so don't worry if you don't get it

me if just means
brainrot